CSET 346 – Natural Language Processing

Project Title: Multimodal NLP for Deepfake Detection and Explainable Content Verification

1. Motivation

Deepfakes are AI-generated or manipulated images, videos, audio, and multimodal content that can imitate real people, events, or speech. Traditional deepfake detectors often focus on a single modality, such as visual artifacts in an image or acoustic artifacts in speech. However, real-world misinformation frequently combines modalities—for example, a manipulated face with synthetic speech, or a genuine video paired with misleading text. Multimodal Natural Language Processing (NLP) provides an opportunity to jointly analyze language, visual, and/or audio evidence. The project will investigate whether combining multimodal representations improves deepfake detection and whether the system can provide an interpretable explanation for its decision.

2. Problem Statement

The primary problem addressed by this project is the reliable detection and explanation of deepfake or manipulated multimodal content using NLP and multimodal learning. Students will develop a multimodal system that analyzes at least two modalities—text plus image/video, or text plus audio—and predicts whether the content is REAL, FAKE, or UNCERTAIN. The system should identify evidence contributing to the decision, such as inconsistent captions, semantic mismatch between text and image, suspicious speech content, or visual/audio manipulation indicators.

Research Question: How effectively can a multimodal NLP model combine textual and non-textual evidence to detect deepfake content, and does multimodal fusion outperform strong single-modality baselines while providing useful explanations?

3. Proposed Scope

The project will create or use a documented dataset containing genuine and manipulated multimodal samples. A minimum of 300 labeled samples is recommended for the course project, with balanced representation of REAL and FAKE classes where feasible. Students should compare single-modality baselines with a multimodal model and perform at least one generalization experiment.

- Use at least two modalities: text + image/video, or text + audio. A three-modality system is encouraged where resources permit.
- Include deepfake/manipulated examples and genuine examples with clearly documented labels and data sources.
- Perform preprocessing appropriate to each modality: text normalization/tokenization, image or video frame sampling, and/or audio feature extraction.
- Use at least two baseline methods, such as text-only classification and visual/audio-only classification.
- Implement a multimodal fusion approach using feature-level, decision-level, or cross-attention fusion.
- Include an explainability component showing which textual, visual, or audio evidence influenced the prediction.
- Perform at least one generalization experiment across manipulation type, dataset/source, language, or modality availability.
- Conduct sanity checks using clearly genuine, clearly manipulated, and deliberately mismatched multimodal pairs.

- Analyze at least 15 difficult cases involving semantic mismatch, identity manipulation, lip-sync inconsistency, synthetic speech, misleading captions, or cross-modal contradictions.
- Address ethical considerations, privacy, consent, dataset bias, and responsible use of deepfake technology.

4. Expected Outcome

The final system should classify each multimodal sample and provide an evidence-based explanation. Students should demonstrate whether multimodal information improves detection compared with single-modality approaches. The project should also identify failure cases, including situations where one modality is authentic but another is manipulated or misleading. The work must provide a meaningful NLP/multimodal extension rather than only applying an existing classifier without analysis.

5. Required Experiments

- Experiment 1: Text-only, image/video-only, or audio-only baseline vs. multimodal model.
- Experiment 2: Compare at least two multimodal fusion strategies or two model architectures.
- Experiment 3: Evaluate generalization to a different manipulation type, data source, language, or unseen modality combination.
- Experiment 4: Perform modality-ablation tests by removing one modality at a time.
- Experiment 5: Conduct qualitative error analysis for at least 15 difficult or incorrectly classified examples.
- Experiment 6: Evaluate explanation quality by checking whether highlighted evidence is relevant to the model's prediction.

6. Evaluation

Performance will be assessed using:

- Accuracy, Precision, Recall, and F1-score.
- ROC-AUC and/or PR-AUC where appropriate.
- Confusion matrix and class-wise performance.
- Separate evaluation for each modality and the multimodal model.
- Ablation results showing the contribution of each modality.
- Cross-dataset or cross-manipulation generalization performance where feasible.
- Explanation evaluation using qualitative assessment and, where applicable, localization/faithfulness measures.

Students must discuss whether multimodal evidence improves robustness over single-modality detection, especially for cases involving conflicting or incomplete information across modalities.

7. Expected Output

For each input sample, the system should ideally produce:

- Prediction: REAL / FAKE / UNCERTAIN
- Confidence score or probability
- Detected modality/manipulation type, where implemented
- Key textual, visual, and/or audio evidence supporting the decision
- Short explanation of why the content was classified as genuine or manipulated

8. Suggested Technical Pipeline

Multimodal Input → Preprocessing → Modality-Specific Representations → Multimodal Fusion → Deepfake Classification → Confidence Estimation → Explainability → Final Verification Report

Students may use transformer-based language models, vision-language models, audio-language models, multimodal encoders, cross-attention, contrastive learning, or parameter-efficient fine-tuning. The choice of architecture should be justified experimentally.

9. Suggested Datasets / Data Sources

Students may select an appropriate public dataset, subject to its license and research-use conditions. Examples include:

- FaceForensics++ for manipulated facial video research.
- DFDC (Deepfake Detection Challenge) dataset for deepfake video detection.
- FakeAVCeleb for multimodal audio-video deepfake research.
- Fakeddit for multimodal misinformation research involving text and images.
- A carefully constructed course dataset using legally reusable public samples, with source, label, modality, and manipulation metadata documented.

Dataset selection must be justified, and students must document licensing, provenance, class balance, preprocessing, and train/validation/test splits. Students should avoid collecting or redistributing private individuals' media without appropriate permission.

10. Recommended Research Extensions

- Multilingual or Hindi-English code-mixed deepfake/misinformation detection.
- Cross-lingual multimodal misinformation detection for Indian languages.
- Explainable multimodal deepfake detection using evidence grounding.
- Audio-visual-text consistency checking for synthetic media.
- Robust detection under compression, low resolution, noisy audio, and social-media transformations.
- Retrieval-Augmented Multimodal Verification using trusted external evidence.
- Deepfake detection with uncertainty estimation and human-in-the-loop verification.

11. Deliverables Checklist

- 1. Source Code (GitHub)
- 2. Research Paper (IEEE Style Word & PDF)
- 3. Presentation Slides
- 4. Dataset description and preprocessing documentation
- 5. Experimental results, confusion matrix, and ablation study
- 6. Demonstration of the multimodal detection system

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Reference / Background Sources

1. Rossler, A., Cozzolino, D., Verdoliva, L., et al. (2019). FaceForensics++: Learning to Detect Manipulated Facial Images. IEEE/CVF International Conference on Computer Vision.

2. Dolhansky, B., Howes, R., Pflaum, B., Baram, N., & Ferrer, C. C. (2020). The DeepFake Detection Challenge Dataset. arXiv:2006.07397.

3. Khalil, M., et al. (2021). FakeAVCeleb: A Novel Audio-Video Multimodal Deepfake Dataset. arXiv:2108.05080.

4. Sharma, K., et al. (2020). Combating Fake News: A Survey on Identification and Mitigation Techniques. ACM Computing Surveys.